

Aditya Verma

B.Tech in Computer Science and Engineering (Data Science)

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SUMMARY

Computer Science undergraduate with strong coding skills (CodeChef 3 ★, Global Rank 34) and practical experience in full-stack development and machine learning. Proficient in designing and deploying scalable web applications using React, Node.js, and MongoDB, with additional expertise in building ML models for real-time applications.

EDUCATION

B.Tech in Computer Science and Engineering (Data Science)
Techno Main Salt Lake, Kolkata, West Bengal

| GPA: 7.58/10 | Expected: May 2026

TECHNICAL SKILLS

- **Languages:** C++, Python, Java, C, SQL
- **Libraries/Frameworks:** NumPy, Pandas, Matplotlib, Scikit-learn, Tensorflow
- **Web Technologies:** HTML, CSS, JavaScript, React.js, Node.js, Express.js, Tailwind CSS
- **Database:** MongoDB, MySQL, PostgreSQL
- **Tools & Platforms:** Git, GitHub, VS Code, Postman, JWT, Tableau
- **Core Concepts:** Data Science and Algorithm, Object Oriented Programming, Operating System, Database Management System, Application Programming Interfaces, Computer Network, System Design

PROJECTS

Sneaker Hub |

[Link](#) | React, Node.js, Express, MongoDB, JWT

- Engineered secure user authentication flows (**JWT, OAuth2.0**) within the e-commerce application, contributing over 6000 lines of code and enabling seamless login experiences for all users.
- Developed and deployed a scalable **REST API using Node.js and MongoDB, cutting API latency by 20%** through optimized database queries and caching strategies, resulting in faster response times.
- **Designed and automated unit & functional test suites**, significantly improving code coverage, reliability, and regression detection.

Facial Emotion Recognition System |

[Link](#) | Python, Tensorflow, OpenCV

- Engineered a real-time facial emotion recognition system leveraging a custom **CNN** architecture trained on the **FER2013 dataset**.
- Optimized model performance via hyperparameter tuning (learning rate schedules, dropout, batch normalization) and data augmentation, achieving **64%** validation accuracy.
- Deployed the solution with OpenCV integration for live webcam inference, enabling interactive, real-time emotion classification.

VAANI(Voice and Audio Analysis Natural Interface) |

[Link](#) | Python, Tensorflow, Conv1D, BiLSTM

- Developed a real-time speech emotion and speaker recognition system using MFCC-based audio features and deep learning models.
- Designed and trained **Conv1D + BiLSTM + Attention networks** to classify human emotions and identify speakers from live microphone input.
- Implemented real-time audio capture, preprocessing, and inference pipeline with noise-robust augmentation and optimized training strategies.
- Achieved a high accuracy of **96% and 89%** respectively using multi-dataset training (RAVDESS, TESS, CREMA-D, SAVEE) and advanced learning rate scheduling techniques.

ACHIVEMENTS

- ❖ Rated 3 ★ on CodeChef with Rating 1616. [CC Profile](#)
- ❖ Rated Pupil on Codeforces. [CC Profile](#)
- ❖ Secured Global Rank 34 in Global Starters 158 on Codechef among 10000+ participants
- ❖ Solved 500+ questions on different coding platforms and online judges

SOCIAL ENGAGEMENTS

Club Membership: Active member of Techno Main Salt Lake's Sport Activity Club.

Sports-Engagements: Cultivated discipline and teamwork through consistent participation in team sports, attending an average of 3 sessions per week over 2 years, enhancing collaborative skills.

Reading: Synthesized insights from books across diverse genres, including biographies and technology, to formulate three novel strategies presented during team brainstorming sessions contributing to increased innovation.